

A Concise Introduction to Seiberg–Witten Theory and Its Applications to Four-Manifolds

Ziang Li

November 2025

Abstract

This report presents a concise introduction to Seiberg–Witten theory and its applications to smooth four-manifolds. We begin with connections, curvature, and Hodge theory on oriented Riemannian manifolds. We then describe spin and Spin^c structures and the Dirac operator. With these tools, we introduce the Seiberg–Witten equations and the compactness and transversality results needed to study their moduli spaces. Finally, we discuss constraints on intersection forms, minimal genera of embedded surfaces, knot surgery, and the relation with symplectic topology. The analytical and topological results whose complete proofs lie beyond this report are stated with their hypotheses and sources.

1 Preliminary

1.1 Connections and Curvature

Let M be a smooth manifold and $E \rightarrow M$ a smooth real or complex vector bundle. Write $\mathfrak{X}(M) = \Gamma(TM)$ for smooth vector fields and $\Gamma(E)$ for smooth sections. Sections in distinct fibers cannot be subtracted directly; a connection makes it possible to differentiate them along vector fields.

Definition 1.1. A connection ∇^A on E is an $\mathbb{R}$ -linear map $\mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E)$, $(X, \sigma) \mapsto \nabla_X^A \sigma$, such that for $f \in C^\infty(M)$,

$$\nabla_X^A(f\sigma) = (Xf)\sigma + f\nabla_X^A\sigma, \quad \nabla_{fX+Y}^A\sigma = f\nabla_X^A\sigma + \nabla_Y^A\sigma.$$

For a complex bundle the operator is also complex-linear in σ .

Equivalently, $d_A : \Gamma(E) \rightarrow \Omega^1(M; E)$ is given by $(d_A\sigma)(X) = \nabla_X^A\sigma$ and satisfies $d_A(f\sigma) = df \otimes \sigma + f d_A\sigma$. Here $\Omega^p(M; E) = \Gamma(\wedge^p T^*M \otimes E)$.

The two descriptions emphasize different uses of the same operator. The first differentiates a section in a specified direction; the second packages all directional derivatives into an E -valued one-form. Evaluating $d_A\sigma$ at X recovers $\nabla_X^A\sigma$. The Leibniz rule is indispensable: when a local frame changes, derivatives of its transition functions must be accounted for. There is no analogous difficulty for an ordinary function, whose values at nearby points lie in the same field.

In a local frame, a section is a column vector and

$$d_A\sigma = d\sigma + \omega\sigma,$$

where ω is a matrix of one-forms. For the trivial bundle the connection d has $\omega = 0$, but a general connection on that same bundle need not. If local coordinates of a section obey $\sigma_\alpha = g_{\alpha\beta}\sigma_\beta$, then on an overlap

$$\omega_\alpha = g_{\alpha\beta}\omega_\beta g_{\alpha\beta}^{-1} + g_{\alpha\beta}d(g_{\alpha\beta}^{-1}).$$

This transformation rule ensures that the local derivatives define the same global connection. A connection compatible with a real Euclidean metric or a Hermitian metric has forms in $\mathfrak{so}(m)$ or $\mathfrak{u}(m)$ in orthonormal frames.

To see the inhomogeneous term directly, write $\sigma_\alpha = g_{\alpha\beta}\sigma_\beta$. Requiring $d\sigma_\alpha + \omega_\alpha\sigma_\alpha = g_{\alpha\beta}(d\sigma_\beta + \omega_\beta\sigma_\beta)$ gives

$$(dg_{\alpha\beta})\sigma_\beta + \omega_\alpha g_{\alpha\beta}\sigma_\beta = g_{\alpha\beta}\omega_\beta\sigma_\beta,$$

which is precisely the transformation rule above. Hermitian compatibility means

$$X\langle\sigma, \tau\rangle = \langle\nabla_X^A\sigma, \tau\rangle + \langle\sigma, \nabla_X^A\tau\rangle;$$

in a unitary frame, $\omega^* = -\omega$.

Proposition 1.2. *If $F : N \rightarrow M$ is smooth, a connection A on $E \rightarrow M$ induces a connection F^*A on $F^*E \rightarrow N$ characterized, for $\sigma \in \Gamma(E)$, by*

$$d_{F^*A}(F^*\sigma) = (dF)^*(d_A\sigma).$$

Here $(dF)^$ pulls back the cotangent factor as well as the bundle factor.*

Proof. In a frame pulled back from a frame of E , the connection form is $F^*\omega$. The local identity is the chain rule, and the transformation rule above makes it independent of the frame. $\square$

Proposition 1.3. *If $F_0, F_1 : N \rightarrow M$ are smoothly homotopic, then F_0^*E and F_1^*E are isomorphic as vector bundles.*

Proof. Pull E back to $N \times [0, 1]$ along the homotopy and use parallel transport along the interval with any connection. The induced isomorphism depends on the chosen transport, but its existence does not. $\square$

The connection extends to $\Omega^p(M; E)$ by the graded Leibniz rule. Its curvature is the bundle endomorphism-valued two-form $F_A = d_A^2 \in \Omega^2(M; \text{End } E)$, or equivalently

$$F_A(X, Y)\sigma = \nabla_X^A \nabla_Y^A \sigma - \nabla_Y^A \nabla_X^A \sigma - \nabla_{[X, Y]}^A \sigma.$$

Locally the structure equation is $F_A = d\omega + \omega \wedge \omega$, and $F_{A,\alpha} = g_{\alpha\beta}F_{A,\beta}g_{\alpha\beta}^{-1}$. These formulas describe the failure of covariant derivatives to commute.

Indeed, applying $d_A = d + \omega$ twice gives

$$(d + \omega)^2\sigma = (d\omega + \omega \wedge \omega)\sigma;$$

the first-order terms cancel by the graded Leibniz rule. Unlike the law for ω , the conjugation law for F_A contains no derivative of $g_{\alpha\beta}$, so curvature is a global tensor. One further covariant derivative gives the Bianchi identity $d_A F_A = 0$. For a line bundle the commutator term vanishes and this reduces to $dF_A = 0$.

For a Hermitian line bundle L , the endomorphisms are scalar and F_A is a globally defined imaginary-valued closed two-form. Locally $d_A = d + i\alpha$ for a real one-form α , so $F_A = i d\alpha$. Throughout this report we retain the imaginary-valued curvature and use the convention

$$c_1(L) = \left[\frac{iF_A}{2\pi} \right] \in H^2(M; \mathbb{R}).$$

The right side is the real image of the integral first Chern class. In particular we will not silently identify F_A with the real two-form $-iF_A$.

The sign follows from $d_A = d + i\alpha$: the displayed representative is $-d\alpha/(2\pi)$ in this frame. A local potential need not be global, so this local exactness does not make $c_1(L)$ vanish. On overlaps the potentials differ by logarithmic derivatives of $U(1)$ transition functions; their periods account for integrality. Later A will always mean a connection on the determinant line of a Spin^c structure unless a different line is explicitly named.

Chern–Weil theory constructs characteristic classes from curvature [11]. For a complex bundle E with a unitary connection,

$$\text{ch}(E) = \text{tr} \exp \left(\frac{iF_A}{2\pi} \right) = \text{rank } E + \text{tr} \frac{iF_A}{2\pi} + \frac{1}{2} \text{tr} \left(\frac{iF_A}{2\pi} \right)^2 + \cdots.$$

Wedge products are understood in the powers. Each component is closed, and its de Rham class is independent of A . The degree-two component is $c_1(E)$; the higher components of $\text{ch}(E)$ are generally rational cohomology classes and are not themselves the higher Chern classes.

In the notation of the original report, put $\tau_k(A) = \text{tr}((iF_A/(2\pi))^k)$. The $2k$ -form $\tau_k(A)$ is closed and $[\tau_k(A)]$ is independent of the connection, but the Chern character has factorial coefficients:

$$\text{ch}(E) = \sum_{k \geq 0} \frac{1}{k!} \tau_k(E), \quad \tau_0(E) = \text{rank } E.$$

To understand independence, join two connections by A_t and differentiate the invariant polynomial in F_{A_t} . The Bianchi identity and cyclicity of trace express that derivative as an exterior derivative. Integrating over t shows that changing the connection changes the representative by an exact form. For a line bundle, $\text{ch}(L) = \exp(c_1(L))$ in rational cohomology. This is the sense in which curvature represents the characteristic class used later in the index formula.

1.2 Classification of Connections

Fix a Hermitian line bundle $L \rightarrow M$ and a reference unitary connection A_0 . Any other unitary connection has the unique form

$$A = A_0 + ia, \quad a \in \Omega^1(M; \mathbb{R}), \quad F_A = F_{A_0} + i da.$$

Thus the space $\mathcal{A}(L)$ is affine over $i\Omega^1(M; \mathbb{R})$. There is no preferred zero connection, although differences of connections are one-forms.

The gauge group $\mathcal{G} = C^\infty(M, S^1)$ acts by multiplication on sections. With the convention that a transformed section is $g\sigma$, its action on connections is

$$g \cdot A = gAg^{-1} = A - (dg)g^{-1}, \quad F_{g \cdot A} = F_A.$$

When $g = e^{iu}$, the real form a changes to $a - du$. On a connected manifold, constant transformations fix every connection. Fix $p_0 \in M$ and put $\mathcal{G}_0 = \{g : g(p_0) = 1\}$. Evaluation at p_0 gives $\mathcal{G} \cong \mathcal{G}_0 \times S^1$, and $\mathcal{G}_0$ acts freely on $\mathcal{A}(L)$. Since constants have no effect on connections,

$$\mathcal{B}(L) := \mathcal{A}(L)/\mathcal{G} \cong \mathcal{A}(L)/\mathcal{G}_0.$$

The distinction between these two quotients will matter when spinors are included: constant gauges then act nontrivially on a nonzero spinor.

The quotient records a geometric connection rather than its expression in one frame. If $A = A_0 + ia$, a gauge identifies it with $A_0 + i(a - (dg)g^{-1}/i)$. The one-form $(dg)g^{-1}/i$ is closed, since $U(1)$ is abelian, but need not be exact when g has nonzero winding. These large gauges are why the quotient can contain a torus even though the affine space of connections is contractible. Based gauges remove the ineffective constant subgroup without changing this quotient.

If M is connected and simply connected, every based map $g : M \rightarrow S^1$ has a unique real lift u with $u(p_0) = 0$. Both $\mathcal{G}_0$ and $\mathcal{A}(L)$ are contractible, and the connection quotient is contractible. More generally, on a connected compact manifold,

$$\pi_0(\mathcal{G}_0) \cong H^1(M; \mathbb{Z}), \quad \pi_1(\mathcal{B}(L)) \cong H^1(M; \mathbb{Z}), \quad \pi_j(\mathcal{B}(L)) = 0 \quad (j > 1).$$

Indeed S^1 is a $K(\mathbb{Z}, 1)$, each component of $\mathcal{G}_0$ is contractible, and $\mathcal{A}(L) \rightarrow \mathcal{B}(L)$ is a principal $\mathcal{G}_0$ -bundle in a suitable Sobolev completion. Consequently the *whole* connected space $\mathcal{B}(L)$ has homotopy type $K(H^1(M; \mathbb{Z}), 1)$; it is not a disjoint union of such spaces.

The calculation also follows from the long exact sequence of $\mathcal{G}_0 \rightarrow \mathcal{A}(L) \rightarrow \mathcal{B}(L)$. Since the middle space is contractible, $\pi_j(\mathcal{B}(L)) \cong \pi_{j-1}(\mathcal{G}_0)$ for $j \geq 1$. Thus disconnectedness of the *gauge group* becomes a fundamental group of the *connection quotient*; it does not make the quotient disconnected. If $b_1 = 0$, the torus below is a point even when M is not simply connected.

1.3 Hodge Theory

From here on, unless otherwise stated, M is a closed (compact without boundary), connected, oriented, smooth Riemannian four-manifold. The metric and orientation determine

$$* : \Omega^p(M) \longrightarrow \Omega^{4-p}(M), \quad *^2 = (-1)^{p(4-p)} = (-1)^p$$

on p -forms. The L^2 product is $(\alpha, \beta) = \int_M \alpha \wedge *\beta$ for real forms, with complex conjugation on the second factor for complex forms. Since M has no boundary, the formal adjoint d^* of d on p -forms is $d^* = (-1)^{4p+5} * d * = - * d *$. We write $\Delta = dd^* + d^*d$.

Definition 1.4. A form $\alpha \in \Omega^p(M)$ is harmonic if $\Delta\alpha = 0$. Its space is denoted $\mathcal{H}^p(M)$.

Theorem 1.5 (Hodge theorem [6]). *Every class in $H^p(M; \mathbb{R})$ has a unique harmonic representative. Moreover,*

$$\Omega^p(M) = \mathcal{H}^p(M) \oplus d\Omega^{p-1}(M) \oplus d^*\Omega^{p+1}(M)$$

orthogonally in the smooth Hodge decomposition.

Integration by parts gives $(\Delta\alpha, \alpha) = \|d\alpha\|^2 + \|d^*\alpha\|^2$, so harmonic forms are precisely the forms that are both closed and coclosed. The existence and decomposition are the elliptic part of the theorem.

For example, on a function f , $d^*f = 0$ and $\Delta f = d^*df$. A harmonic function on connected closed M is constant, hence $b_0 = 1$. Closedness and coclosedness give the easy direction of Hodge theory;

obtaining the representative of each de Rham class uses the elliptic decomposition. This matters because the self-dual harmonic forms below measure a finite-dimensional obstruction to a differential equation.

On two-forms $*^2 = 1$. Write $\Omega_{\pm}^2(M)$ for the ± 1 eigenspaces and $P_{\pm} = (1 \pm *)/2$ for their projections. Thus

$$\alpha = \alpha^+ + \alpha^-, \quad \alpha^{\pm} = P_{\pm}\alpha.$$

As $*$ commutes with Δ , the harmonic two-forms split into $\mathcal{H}_+^2 \oplus \mathcal{H}_-^2$. Their dimensions b_+ and b_- are the positive and negative indices of the intersection form on $H^2(M; \mathbb{R})$; its signature is $\sigma(M) = b_+ - b_-$.

For harmonic real two-forms, $\int_M \alpha \wedge \beta = (\alpha, * \beta)$. Thus this pairing is positive definite on $\mathcal{H}_+^2$, negative definite on $\mathcal{H}_-^2$, and the two summands are orthogonal. Their dimensions are metric independent, although the particular harmonic subspaces generally depend on the metric. We use $b_1 = \dim H^1(M; \mathbb{R})$ and $\chi(M) = 2 - 2b_1 + b_+ + b_-$ throughout.

The operator $d^+ = P_+ d : \Omega^1(M) \rightarrow \Omega_+^2(M)$ occurs in the elliptic complex

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d^+} \Omega_+^2(M) \longrightarrow 0.$$

Its cohomology is $\mathcal{H}^0$, $\mathcal{H}^1$, and $\mathcal{H}_+^2$. Equivalently, $d^* \oplus d^+$ is elliptic, with kernel $\mathcal{H}^1$ and cokernel $\mathcal{H}^0 \oplus \mathcal{H}_+^2$. This will control both gauge fixing and the dimension of the monopole moduli space.

For a fixed Hermitian line bundle, the self-dual curvatures of unitary connections form the affine subspace

$$F_{A_0}^+ + i d^+ \Omega^1(M; \mathbb{R}) \subset i \Omega_+^2(M).$$

It has codimension b_+ : the quotient of Ω_+^2 by $d^+ \Omega^1$ is $\mathcal{H}_+^2$. This gives the obstruction to solving a prescribed self-dual curvature equation.

To solve $F_A^+ = \eta$, write $A = A_0 + ia$ and ask for $id^+ a = \eta - F_{A_0}^+$. Projecting to $\mathcal{H}_+^2$ gives the necessary condition

$$\text{pr}_{\mathcal{H}_+^2}(\eta - F_{A_0}^+) = 0.$$

Hodge theory for the elliptic complex makes it sufficient. A generic perturbation therefore avoids reducible monopoles when $b_+ > 0$, since a reducible has $\psi = 0$ and obeys just this curvature equation. If $b_+ = 0$, this harmonic obstruction disappears and a reducible cannot be removed in this way.

There is also a useful description of the entire connection quotient. Define

$$\mathcal{C}_L = \left\{ F \in i \Omega^2(M) : dF = 0, \left[\frac{iF}{2\pi} \right] = c_1(L)_{\mathbb{R}} \right\}.$$

Proposition 1.6. *The curvature map $\mathcal{B}(L) \rightarrow \mathcal{C}_L$, $[A] \mapsto F_A$, is onto. Its fibers are tori*

$$H^1(M; \mathbb{R}) / 2\pi H^1(M; \mathbb{Z}).$$

In particular, if $H^1(M; \mathbb{R}) = 0$ it is a bijection. After choosing a reference connection and a splitting, $\mathcal{B}(L)$ is (noncanonically) a product of $\mathcal{C}_L$ and $(S^1)^{b_1}$.

Proof. If $F - F_{A_0} = i da$, then $A_0 + ia$ has curvature F . Two connections have the same curvature exactly when their difference is i times a closed one-form. The one-forms $(dg)g^{-1}/i$ obtained from gauges have periods in $2\pi\mathbb{Z}$; modulo them the closed forms give the stated torus. $\square$

Finally, on line-bundle connections the Yang–Mills functional can be written with the real curvature $f_A = -iF_A$ as

$$\mathcal{Y}(A) = \int_M f_A \wedge *f_A = \|F_A\|_{L^2}^2.$$

The variation $A_t = A + ita$ gives $\frac{d}{dt}\mathcal{Y}(A_t)|_{t=0} = 2\langle d^*f_A, a \rangle$. Its critical points satisfy $d^*F_A = 0$ and, because $dF_A = 0$, have harmonic curvature. In gauge theory an (anti-)self-dual connection obeys the stronger equation $F_A^\mp = 0$; a general abelian Yang–Mills solution need not be an instanton.

The calculation is explicit: $f_{A_t} = f_A + t da$, whence

$$\left. \frac{d}{dt}\mathcal{Y}(A_t) \right|_{t=0} = 2 \int_M \langle f_A, da \rangle d\text{vol} = 2 \int_M \langle d^*f_A, a \rangle d\text{vol}.$$

There is no boundary term because M is closed. The Euler–Lagrange equation is $d^*f_A = 0$, equivalently $d(*f_A) = 0$; with $df_A = 0$ it says $\Delta f_A = 0$. Conversely, $F_A^+ = 0$ or $F_A^- = 0$ implies the Yang–Mills equation by Bianchi. This is a first example of a curvature equation selecting a finite-dimensional geometric locus inside the infinite-dimensional connection space.

2 Spin Geometry and Dirac Operator

2.1 Spin Manifolds

The oriented orthonormal frame bundle of TM has structure group $SO(4)$. Spinors require a lift of this group, or of its extension by $U(1)$. We first describe the groups in dimension four [10].

Identify $\mathbb{R}^4$ with the quaternions, represented by matrices

$$Q = \begin{pmatrix} t + iz & -x + iy \\ x + iy & t - iz \end{pmatrix}, \quad \det Q = t^2 + x^2 + y^2 + z^2.$$

The matrices of norm one form $SU(2) \cong S^3$. The action

$$\varpi : SU(2)_+ \times SU(2)_- \longrightarrow SO(4), \quad \varpi(A_+, A_-)(Q) = A_+ Q A_-^{-1}$$

is onto and has kernel $\{(1, 1), (-1, -1)\}$. Thus $\text{Spin}(4) \cong SU(2)_+ \times SU(2)_-$ is the double cover of $SO(4)$, and $SO(4) \cong (SU(2) \times SU(2))/\{\pm(1, 1)\}$. Rotations of two orthogonal planes can be read from its maximal torus, explaining why two $SU(2)$ factors occur.

For clarity, A_+ and A_- are unit quaternions acting from the left and right. Their action preserves $\det Q$ and hence the Euclidean norm. Both $(1, 1)$ and $(-1, -1)$ act trivially; these are all the elements in the kernel. The image is a connected six-dimensional subgroup of the six-dimensional group $SO(4)$, hence is all of it. The order of the two factors is a convention: reversing it interchanges the labels $+$ and $-$, but the chosen labels must agree with the chirality convention used in the Seiberg–Witten equations.

Definition 2.1. A spin structure on M is a principal $\text{Spin}(4)$ -bundle P_{Spin} together with a two-to-one bundle map to the oriented orthonormal frame bundle P_{SO} intertwining ϖ .

Equivalently, it is a compatible lift of the transition functions. The failure of locally chosen lifts to obey the cocycle condition is measured by $w_2(TM)$. Therefore M is spin if and only if $w_2(TM) = 0$. The standard complex two-dimensional representations of the two $SU(2)$ factors yield spinor bundles S^+ and S^- of complex rank two.

More concretely, if $h_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow SO(4)$ are frame transition functions, choose lifts $\tilde{h}_{\alpha\beta}$ on a suitable cover. On a triple overlap their product may be 1 or -1 in the kernel. These signs form a $\mathbb{Z}_2$ -valued two-cocycle representing $w_2(TM)$. Vanishing of its class lets us adjust the lifts to satisfy the cocycle condition and glue a principal $\text{Spin}(4)$ -bundle. Once such a lift exists, $S^\pm = P_{\text{Spin}} \times_{SU(2)_\pm} \mathbb{C}^2$. It is the associated bundles, not the bare representations, that carry the spinor fields used by a Dirac operator.

A manifold need not be spin. The group

$$\text{Spin}^c(4) = (\text{Spin}(4) \times U(1))/\{(1, 1), (-1, -1)\}$$

fits into two useful exact sequences,

$$1 \rightarrow U(1) \rightarrow \text{Spin}^c(4) \rightarrow SO(4) \rightarrow 1, \quad 1 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}^c(4) \xrightarrow{(\varpi, \det)} SO(4) \times U(1) \rightarrow 1,$$

where $\det([s, z]) = z^2$. A Spin^c structure is a lift $P_{\mathfrak{s}} \rightarrow P_{SO}$ through the first projection. It has rank-two positive and negative complex spinor bundles W^+ and W^- , and a *determinant line* $\mathcal{L}$ with

$$\det W^+ \cong \det W^- \cong \mathcal{L}, \quad c_1(\mathcal{L}) \equiv w_2(TM) \pmod{2}.$$

The symbols $W^\pm$ denote globally defined Spin^c spinor bundles. Only when a spin structure and an auxiliary line E are specified may one write $W^\pm = S^\pm \otimes E$; then $\mathcal{L} = E^2$. There is no need for $\mathcal{L}$ to have a global square root.

The determinant map squares the $U(1)$ factor because scalar multiplication by z on a complex rank-two spinor fiber has determinant z^2 . The reduction of $c_1(\mathcal{L})$ modulo two cancels the obstruction of the spin lift. This explains both the congruence $c_1(\mathcal{L}) \equiv w_2(TM)$ and why the determinant line alone need not specify the entire Spin^c structure. The notation $S^\pm \otimes E$ is convenient in a local spin trivialization; it is generally invalid globally when the manifold is not spin.

Theorem 2.2 (Existence of Spin^c structures [10]). *Every closed oriented smooth four-manifold has a Spin^c structure. On an oriented manifold of any dimension, a Spin^c structure exists precisely when $w_2(TM)$ lifts to an integral class.*

For dimension four the integral lift exists (equivalently the integral third Stiefel–Whitney obstruction vanishes). Twisting a Spin^c structure by a line bundle E replaces $W^\pm$ by $W^\pm \otimes E$ and $\mathcal{L}$ by $\mathcal{L} \otimes E^2$. Isomorphism classes of Spin^c structures form a torsor for $H^2(M; \mathbb{Z})$. These facts will explain why the monopole invariants are indexed by Spin^c structures rather than only by determinant classes.

There is no preferred origin in this torsor. If $\mathfrak{s}_0$ is chosen, however, each other structure has a unique expression $\mathfrak{s}_0 \otimes E$ up to isomorphism of E , and

$$c_1(\det(\mathfrak{s}_0 \otimes E)) = c_1(\det \mathfrak{s}_0) + 2c_1(E).$$

If $H^2(M; \mathbb{Z})$ has two-torsion, the doubling map need not be injective, so two distinct structures can have the same determinant Chern class. We will retain $\mathfrak{s}$ as the index of a moduli space and use the characteristic class only when it is adequate to specify the structure.

2.2 Spin Connection

The Levi–Civita connection is the unique metric-compatible, torsion-free connection on TM . When M is spin, it lifts uniquely to P_{Spin} and gives connections on S^+ and S^- .

The uniqueness here is for a *fixed* spin structure and a fixed Levi–Civita connection; it does not mean that the manifold has a unique spin structure. In local oriented orthonormal frames the Levi–Civita connection has a skew-symmetric matrix of one-forms ω_{ij} . Lifting its Lie-algebra-valued form through the double cover and applying the spin representation produces the local spinor connection below. The analogous construction for a Spin^c structure requires one extra choice, a connection on its determinant line.

For a tangent vector v , Clifford multiplication $\rho(v) : W^\pm \rightarrow W^\mp$ satisfies

$$\rho(v)\rho(w) + \rho(w)\rho(v) = -2g(v, w) \text{id}.$$

At each point the complexified Clifford algebra $\text{Cl}(T_p^*M) \otimes \mathbb{C}$ is isomorphic to $\text{End}_{\mathbb{C}}(W_p^+ \oplus W_p^-)$; as vector spaces it is identified with $\Lambda^* T_p^*M \otimes \mathbb{C}$. This is a vector-space identification, not an identification of exterior and Clifford products. Two-forms act within the two chiral summands; with our choice of chirality, a self-dual two-form acts on W^+ , while an anti-self-dual one acts on W^- . This is why the curvature equation below uses F_A^+ and W^+ .

For an orthonormal coframe $e^1, \dots, e^4$, the relations read $\rho(e^j)^2 = -1$ and $\rho(e^i)\rho(e^j) = -\rho(e^j)\rho(e^i)$ for $i \neq j$. The grading $W = W^+ \oplus W^-$ is reversed by a one-form and preserved by an even Clifford product. The decomposition $\Lambda^2 = \Lambda_+^2 \oplus \Lambda_-^2$ from Hodge theory therefore becomes a decomposition of the two-form part of the action on the two chiral summands. In particular, $\rho : i\Lambda_+^2 \rightarrow \text{Herm}_0(W^+)$

is a real-linear isomorphism fiberwise, which will let us turn a quadratic spinor endomorphism into a curvature two-form.

In an orthonormal frame $e_1, \dots, e_4$, if $\omega_{ij} = -\omega_{ji}$ are the Levi-Civita connection one-forms, the lifted spin connection has the local expression

$$\nabla^{\text{spin}} = d + \frac{1}{4} \sum_{i,j} \omega_{ij} \rho(e_i) \rho(e_j).$$

The coefficient $1/4$ follows from the double-cover map on Lie algebras. Its defining compatibility is

$$\nabla_X(\rho(\alpha)\psi) = \rho(\nabla_X^{LC} \alpha)\psi + \rho(\alpha)\nabla_X\psi$$

for one-forms α and spinors ψ .

This compatibility is also the appropriate replacement for the informal local definition that a spin connection merely has a Lie-algebra-valued form. If α is changed as a section of T^*M , both its ordinary derivative and the connection on the spinors change; the two terms on the right ensure that Clifford multiplication remains parallel. It is the property used to derive the Dirac operator and its square without dependence on a particular frame.

Theorem 2.3. *For a fixed Spin^c structure and a unitary connection A on its determinant line $\mathcal{L}$, there is a unique Spin^c connection ∇^A on $W^+ \oplus W^-$ that induces the Levi-Civita connection on TM and A on $\mathcal{L}$ [10, 12].*

Locally $W^\pm = S^\pm \otimes E$, where $E^2 \cong \mathcal{L}$, and the connection is the tensor product of the lifted spin connection and the local connection on E whose doubled connection is A . If A changes by ia , then ∇^A changes by $ia/2$ on spinors. This factor is essential in the Dirac and gauge-action formulas.

For example, if locally $A = d + ia$ on $\mathcal{L}$, its contribution to the spinor connection is $ia/2$. Two such local descriptions agree on overlaps because the Spin^c transition functions simultaneously change the spin and circle factors. Thus the local square-root description is a calculation device rather than an assertion that E exists globally. The uniqueness statement makes it meaningful to write D_A using only the globally defined determinant connection A .

2.3 The Dirac Operator

Definition 2.4. For the Spin^c connection ∇^A , the Dirac operator is

$$D_A = \sum_{j=1}^4 \rho(e^j) \nabla_{e_j}^A : \Gamma(W^+ \oplus W^-) \longrightarrow \Gamma(W^+ \oplus W^-),$$

where e_j is a local orthonormal frame and e^j its dual. Its chiral part is $D_A^+ : \Gamma(W^+) \rightarrow \Gamma(W^-)$; the other part $D_A^- : \Gamma(W^-) \rightarrow \Gamma(W^+)$ is its formal adjoint on closed M .

The expression is independent of the orthonormal frame. On flat $\mathbb{R}^4$ with the trivial connection, $D = \sum_j \rho(dx^j) \partial_{x_j}$ and $D^2 = -\sum_j \partial_{x_j}^2$ by the Clifford relations. More generally, if $A' = A + ia$ then

$$D_{A'} = D_A + \frac{i}{2} \rho(a).$$

The flat calculation can be written out as

$$D^2 = \sum_{i,j} \rho(dx^i) \rho(dx^j) \partial_i \partial_j = \sum_i \rho(dx^i)^2 \partial_i^2 + \sum_{i < j} (\rho(dx^i) \rho(dx^j) + \rho(dx^j) \rho(dx^i)) \partial_i \partial_j = -\sum_i \partial_i^2.$$

The sign agrees with the nonnegative Laplacian d^*d on functions. On curved M , covariant derivatives do not commute, and their commutator produces exactly the curvature correction in the Weitzenböck formula. The displayed change-of-connection formula is another direct consequence of the $1/2$ factor in the Spin^c lift.

There is a related Dirac-type operator on differential forms: $d + d^*$, for the Clifford action $\alpha \mapsto \alpha \wedge -\iota_{\alpha^\sharp}$. Its square is Δ , so its kernel consists of harmonic forms. On a closed four-manifold it interchanges even and odd forms,

$$(d + d^*)^{\text{ev}} : \Omega^0 \oplus \Omega^2 \oplus \Omega^4 \longrightarrow \Omega^1 \oplus \Omega^3,$$

and its real index is $\sum_p (-1)^p b_p = \chi(M)$. This example shows how an analytic index can be topological.

Indeed, $(d + d^*)^2 = dd^* + d^*d = \Delta$. If $(d + d^*)\omega = 0$, taking the L^2 product with $(d + d^*)^2\omega$ shows that $d\omega = d^*\omega = 0$; conversely any harmonic form lies in the kernel. The even kernel has dimension $b_0 + b_2 + b_4$ and the odd kernel has dimension $b_1 + b_3$. Poincaré duality gives $b_4 = b_0 = 1$ and $b_3 = b_1$ here, reproducing the Euler characteristic formula used later. This is a different Dirac-type operator from D_A^+ , but it illustrates why an index theorem can turn a differential operator into a topological number.

Theorem 2.5 (Weitzenböck formula [10, 12]). *For $\psi \in \Gamma(W^+)$, scalar curvature s , and imaginary-valued curvature F_A of the determinant connection,*

$$D_A^- D_A^+ \psi = (\nabla^A)^* \nabla^A \psi + \frac{s}{4} \psi + \frac{1}{2} \rho(F_A^+) \psi.$$

Here ρ extends to two-forms using $\rho(\alpha \wedge \beta) = \frac{1}{2}(\rho(\alpha)\rho(\beta) - \rho(\beta)\rho(\alpha))$, and the final term is a Hermitian endomorphism of W^+ .

The formula connects the curvature equation to pointwise estimates for spinors. This connection is the main ingredient in compactness and positive-scalar-curvature vanishing below.

Taking the L^2 product of the formula with ψ and using $D_A^+ \psi = 0$ gives

$$0 = \int_M \left(|\nabla^A \psi|^2 + \frac{s}{4} |\psi|^2 + \frac{1}{2} \langle \rho(F_A^+) \psi, \psi \rangle \right) d \text{vol}.$$

This is not a formula for the sum-of-squares functional of an arbitrary pair. It holds for a Dirac solution and becomes especially useful after the curvature equation replaces F_A^+ by a quadratic expression in ψ and a fixed perturbation.

Definition 2.6. The complex index of the chiral Dirac operator is $\text{ind}_{\mathbb{C}} D_A^+ = \dim_{\mathbb{C}} \ker D_A^+ - \dim_{\mathbb{C}} \ker D_A^-$.

Theorem 2.7 (Atiyah–Singer index formula [1, 10]). *On a closed oriented Spin^c four-manifold, with $c = c_1(\mathcal{L})$,*

$$\text{ind}_{\mathbb{C}} D_A^+ = \left\langle e^{c/2} \widehat{A}(TM), [M] \right\rangle = \frac{\langle c^2, [M] \rangle - \sigma(M)}{8}.$$

Indeed the degree-four term of $e^{c/2} \widehat{A}$ is $c^2/8 - p_1(TM)/24$, and $\langle p_1(TM), [M] \rangle = 3\sigma(M)$. This index is an integer independent of A and will enter the expected dimension of the moduli space.

To spell out the degree count, $e^{c/2} = 1 + c/2 + c^2/8 + \dots$ and $\widehat{A}(TM) = 1 - p_1(TM)/24 + \dots$. Since M has dimension four, only their degree-four part contributes to pairing with $[M]$:

$$\left\langle \frac{c^2}{8} - \frac{p_1(TM)}{24}, [M] \right\rangle = \frac{c^2}{8} - \frac{\sigma(M)}{8}.$$

Here and below c^2 abbreviates $\langle c \smile c, [M] \rangle$. The complex index is twice its value when regarded as a real index. That distinction will be essential when the real gauge-theoretic deformation complex is combined with the complex Dirac operator.

3 Seiberg–Witten Theory

3.1 Seiberg–Witten Equations and the Moduli Space

Let M be as above, with a fixed Spin^c structure $\mathfrak{s}$, spinor bundles $W^\pm$, and determinant line $\mathcal{L}$. A configuration consists of a unitary connection A on $\mathcal{L}$ and a positive spinor $\psi \in \Gamma(W^+)$. Witten described the four-dimensional equations as involving “an abelian gauge group” [17, p. 769]; the linear Dirac equation and a quadratic curvature equation together produce a finite-dimensional moduli problem.

For $\psi \in W^+$, let $\psi\psi^* : v \mapsto \langle v, \psi \rangle \psi$ and let $(\psi\psi^*)_0 = \psi\psi^* - (|\psi|^2/2) \text{id}_{W^+}$. Clifford multiplication identifies $i\Lambda_+^2 T^*M$ with the traceless Hermitian endomorphisms of W^+ . Define the quadratic map $q : W^+ \rightarrow i\Lambda_+^2 T^*M$ by

$$\rho(q(\psi)) = (\psi\psi^*)_0.$$

In particular $\langle \rho(q(\psi))\psi, \psi \rangle = |\psi|^4/2$. Unlike the original untyped local expression, this definition specifies both the target and the normalization.

The rank-one endomorphism $\psi\psi^*$ has trace $|\psi|^2$ on the rank-two bundle W^+ . Subtracting half its trace makes it traceless, so it belongs to the image of self-dual imaginary two-forms. Applying it to ψ yields $(\psi\psi^*)_0\psi = (|\psi|^2/2)\psi$, proving the identity above. The map is quadratic in the precise sense $q(z\psi) = |z|^2 q(\psi)$. Hence $q(g\psi) = q(\psi)$ for a circle-valued gauge g , as required for gauge invariance.

Definition 3.1. For a fixed smooth imaginary-valued self-dual two-form $\eta \in i\Omega_+^2(M)$, the perturbed Seiberg–Witten equations are

$$D_A^+ \psi = 0, \quad F_A^+ = q(\psi) + \eta.$$

The unperturbed equations have $\eta = 0$.

The first equation is a spinor equation; the second equates two sections of $i\Lambda_+^2 T^*M$. The perturbation helps arrange smoothness without changing the underlying Spin^c structure [17, 12]. One may view solutions as zeroes of the nonnegative gauge-invariant functional

$$\mathcal{S}_\eta(A, \psi) = \int_M (|D_A^+ \psi|^2 + |F_A^+ - q(\psi) - \eta|^2) d\text{vol}.$$

Its *absolute zeroes*, when present, are precisely the solutions. No identity without curvature cross terms is asserted for arbitrary configurations. For a solution, the integrated Weitzenböck formula instead gives the exact relation

$$0 = \int_M \left(|\nabla^A \psi|^2 + \frac{s}{4} |\psi|^2 + \frac{1}{4} |\psi|^4 + \frac{1}{2} \langle \rho(\eta)\psi, \psi \rangle \right) d\text{vol}.$$

When $\eta = 0$ and $s > 0$ everywhere, this forces $\psi = 0$; for suitable generic perturbations and $b_+ > 0$, reducibles can then be excluded. The resulting vanishing of the invariant will be discussed below.

The zero-set formulation explains the coupling without asserting that all critical points of $\mathcal{S}_\eta$ solve the equations: a nonnegative functional can have nonzero critical values. Expanding its second square produces $-2\langle F_A^+, q(\psi) + \eta \rangle$, so the unperturbed expression in the original report omitted terms. On an actual solution, substituting $\rho(q(\psi)) = (\psi\psi^*)_0$ into the preceding integrated formula gives the quartic term $|\psi|^4/4$. If s has a positive lower bound and $\eta = 0$, every term is nonnegative and the spinor vanishes. This leaves the curvature equation $F_A^+ = 0$, whose solvability depends on the Hodge-theoretic harmonic obstruction.

Let $\mathcal{C}_s = \mathcal{A}(\mathcal{L}) \times \Gamma(W^+)$ be the configuration space (distinct from the curvature space $\mathcal{C}_L$ above). A gauge $g : M \rightarrow S^1$ acts on spinors with weight one and on the determinant connection with weight two:

$$g \cdot (A, \psi) = (A - 2(dg)g^{-1}, g\psi).$$

Indeed $D_{A-2(dg)g^{-1}}^+(g\psi) = gD_A^+\psi$. Writing $A = A_0 + ia$ gives $a \mapsto a - 2du$ for $g = e^{iu}$. The factor two is the effect of $\mathcal{L} = \det W^+$ and differs from the action on a line-bundle connection alone in Section 1.2.

One can check the factor in a local spinor frame: differentiating $g\psi$ adds $(dg)\psi$, while changing A by $-2(dg)g^{-1}$ changes ∇^A by $-(dg)g^{-1}$ on spinors. These terms cancel. Curvature is unchanged and $q(g\psi) = q(\psi)$, so both equations descend to the quotient. The configuration space itself is an affine space times a complex vector space, but its quotient can have stabilizers.

Fix $p_0 \in M$ and define $\mathcal{G}_0 = \{g : g(p_0) = 1\}$. The based quotient $\tilde{\mathcal{B}} = \mathcal{C}_s/\mathcal{G}_0$ retains a residual $S^1 = \mathcal{G}/\mathcal{G}_0$ action. A configuration is *irreducible* if ψ is not identically zero; its stabilizer in $\mathcal{G}$ is trivial. For $\psi = 0$ the stabilizer is the constant S^1 , and the configuration is *reducible*. On the irreducible locus $\tilde{\mathcal{B}}^* \rightarrow \mathcal{B}^* = \mathcal{C}_s^*/\mathcal{G}$ is a principal circle bundle. After Sobolev completion, $\mathcal{B}^*$ is a smooth Banach manifold, whereas the full quotient can be singular at reducibles.

If M is simply connected, a gauge can put a in Coulomb gauge $d^*a = 0$. A unique representative requires in addition a normalization of the remaining constant gauge (and, for pairs, no constant gauge normalizes every spinor canonically). Without simple connectivity, harmonic one-forms contribute a compact torus, as in Section 1.2.

In fact solve $\Delta_0 u = \frac{1}{2}d^*a$ with $\int_M u \, d\text{vol} = 0$, and apply $g = e^{iu}$. Then $a' = a - 2du$ has $d^*a' = 0$. On connected M the normalized solution u is unique, but constants still act on ψ and yield the residual circle on the based quotient. If $b_1 > 0$, the co-closed form can also have a harmonic component. Large gauges shift that component by a lattice; in the compactness proof we choose a compact fundamental domain.

Definition 3.2. The moduli spaces for $(M, \mathfrak{s}, \eta)$ are

$$\begin{aligned} \mathcal{M}_{\mathfrak{s}, \eta} &= \{(A, \psi) \in \mathcal{C}_s : D_A^+\psi = 0, F_A^+ = q(\psi) + \eta\}/\mathcal{G}, \\ \widetilde{\mathcal{M}}_{\mathfrak{s}, \eta} &= \{(A, \psi) \text{ solving the same equations}\}/\mathcal{G}_0. \end{aligned}$$

A superscript $*$ restricts either space to irreducible solutions.

The choice of $\mathfrak{s}$, not merely $c_1(\mathcal{L})$ when there is two-torsion, is part of the data of these spaces.

The unperturbed space is the case $\eta = 0$. A class $[A, \psi]$ is reducible exactly when $\psi = 0$, and its stabilizer is a copy of S^1 ; an irreducible solution has trivial stabilizer. Thus the full quotient may have cone-type singularities even when its irreducible part is smooth. This distinction will recur in the proof sketch of Donaldson's theorem, where the local link of a reducible is central to the argument.

3.2 Compactness and Transversality

Compactness makes a count or intersection number finite; transversality gives the expected smooth dimension. Both statements concern gauge-equivalence classes, and the analytic details require Sobolev completions and elliptic regularity [12].

Lemma 3.3 (Spinor bound). *Let (A, ψ) solve the perturbed equations on closed M . If $|\psi|^2$ is maximal at p , then either $\psi(p) = 0$ or*

$$|\psi(p)|^2 \leq -s(p) + 2\|\rho(\eta(p))\|_{\text{op}}.$$

In particular $\|\psi\|_{L^\infty}$ is bounded in terms of the lower scalar-curvature bound and η . If $\eta = 0$ and $s > 0$, then $\psi = 0$.

Proof. Use the convention $\Delta_0 = d^*d \geq 0$ on functions. At a maximum, $\Delta_0|\psi|^2 \geq 0$ and

$$\frac{1}{2}\Delta_0|\psi|^2 = \langle (\nabla^A)^* \nabla^A \psi, \psi \rangle - |\nabla^A \psi|^2.$$

Insert the Weitzenböck formula and $F_A^+ = q(\psi) + \eta$. At the maximum one obtains

$$0 \geq |\nabla^A \psi|^2 + \frac{s}{4}|\psi|^2 + \frac{1}{4}|\psi|^4 + \frac{1}{2}\langle \rho(\eta)\psi, \psi \rangle.$$

Bounding the last term below by $-\frac{1}{2}\|\rho(\eta)\|_{\text{op}}|\psi|^2$ and dividing by $|\psi|^2/4$ proves the claim.

The maximum principle is applied to the scalar function $|\psi|^2$, not to the spinor directly. Replacing $s(p)$ and $\|\rho(\eta(p))\|$ by uniform bounds yields one estimate for every solution. The coefficients depend on our normalization of q , which is why the constants in the original attempted pointwise bound cannot simply be carried over. $\square$

Theorem 3.4 (Compactness [12]). *For a fixed Spin^c structure, Riemannian metric, and smooth perturbation η on a closed four-manifold, $\mathcal{M}_{s,\eta}$ is compact in the smooth topology modulo gauge.*

Sketch of proof. The spinor bound controls $q(\psi)$ and hence $F_A^+ = q(\psi) + \eta$ in L^∞ . The Chern class fixes $\int_M F_A \wedge F_A$, and the identities

$$\int_M F_A \wedge F_A = -\|F_A^+\|_{L^2}^2 + \|F_A^-\|_{L^2}^2$$

for imaginary-valued F_A give an L^2 bound for the full curvature. (Equivalently use the real form $-iF_A$ with the corresponding sign.) After gauge fixing relative to A_0 , the elliptic estimate for $d^* \oplus d^+$ controls the connection one-form modulo its harmonic part. Because a spinor gauge acts on a with weight two, large gauges put that part in a compact fundamental domain of $H^1(M; \mathbb{R})/4\pi H^1(M; \mathbb{Z})$. This differs from the 2π lattice for connections alone in Section 1.2. If M is simply connected, the harmonic part vanishes.

More precisely, Chern–Weil theory gives $\int_M F_A \wedge F_A = -(2\pi)^2 c_1(\mathcal{L})^2$, independent of A . With our imaginary-valued convention the self-dual decomposition therefore yields

$$\|F_A^-\|_{L^2}^2 = \|F_A^+\|_{L^2}^2 - (2\pi)^2 c_1(\mathcal{L})^2.$$

The right side is bounded because F_A^+ is bounded and the topological term is fixed. For a in Coulomb gauge, a representative elliptic estimate is

$$\|a - h\|_{L_1^2} \leq C(\|d^+ a\|_{L^2} + \|d^* a\|_{L^2}),$$

where h is the harmonic one-form component. The estimate does not control h , which is why the compact fundamental domain is needed. If $b_1 = 0$, one may take $h = 0$.

In Coulomb gauge, $A = A_0 + ia$ and the equations take the schematic but correctly normalized form

$$D_{A_0}^+ \psi + \frac{i}{2}\rho(a)\psi = 0, \quad i d^+ a = q(\psi) + \eta - F_{A_0}^+, \quad d^* a = 0.$$

Sobolev multiplication, elliptic estimates for this coupled system, and bootstrapping give uniform bounds for all derivatives after suitable gauges. Rellich compactness followed by elliptic regularity yields a smoothly convergent subsequence. The standard gauge-fixing and bootstrap estimates are the technical input supplied by the cited compactness theorem; the L^∞ bound alone does not imply all higher estimates.

Here $D_{A_0}^+$ and $d^* \oplus d^+$ are elliptic. Once a has a first Sobolev bound, the product $\rho(a)\psi$ is controlled using the uniform bound on ψ and Sobolev multiplication. The Dirac estimate improves the regularity of ψ ; the quadratic map $q(\psi)$ then improves that of a through the second equation. Iterating alternates between these estimates. For a sequence, one obtains bounds in L_k^p for every fixed k and suitable $p > 4$, after compatible gauges. Sobolev embedding controls continuous derivatives, and Rellich compactness gives a convergent subsequence at each stage. A diagonal subsequence and elliptic regularity identify a smooth limiting solution. These are the stages of the original compactness argument; the gauge-fixing and multiplication estimates are the substantial cited analytic input. $\square$

Definition 3.5. A bounded operator $T : E_1 \rightarrow E_2$ between Banach spaces is Fredholm if its kernel and cokernel are finite-dimensional and its image is closed. Its index is $\text{ind } T = \dim \ker T - \dim \text{coker } T$. For Hilbert-space Fredholm operators, $\text{coker } T \cong \ker T^*$; this last formula requires an adjoint in the appropriate spaces.

Finite-dimensional kernel means there are only finitely many infinitesimal solutions to the homogeneous linearized equation. Finite-dimensional cokernel measures the independent obstructions to solving its inhomogeneous version. For an elliptic operator on a closed manifold, elliptic estimates and compact Sobolev embedding supply these finiteness properties in the appropriate completions. The index is stable under lower-order perturbations, which lets us compute the nonlinear deformation problem from its principal elliptic part. At an irreducible solution, linearizing the equations and quotienting infinitesimal gauge transformations gives an elliptic deformation complex. Up to zero-order terms, its real Fredholm operator is the sum of

$$d^* \oplus d^+ : \Omega^1(M; i\mathbb{R}) \longrightarrow \Omega^0(M; i\mathbb{R}) \oplus \Omega_+^2(M; i\mathbb{R})$$

and $D_A^+ : \Gamma(W^+) \rightarrow \Gamma(W^-)$. The first has real index $b_1 - b_+ - 1$ (because M is connected), and the second has real index $2 \text{ind}_{\mathbb{C}} D_A^+$. Hence the expected dimension is

$$d(\mathfrak{s}) = 2 \text{ind}_{\mathbb{C}} D_A^+ - (1 - b_1 + b_+) = \frac{c_1(\mathcal{L})^2 - (2\chi(M) + 3\sigma(M))}{4}. \quad (1)$$

Here $c^2 = \langle c \smile c, [M] \rangle$ and $\chi(M) = 2 - 2b_1 + b_+ + b_-$. In the based quotient the expected dimension is $d(\mathfrak{s}) + 1$. The index need not be positive to state this formula; a negative expected dimension generically means no irreducible solutions.

To verify the last equality rather than merely quote it, use $2 \text{ind}_{\mathbb{C}} D_A^+ = (c^2 - \sigma)/4$ and

$$4(1 - b_1 + b_+) = 2\chi(M) + 2\sigma(M).$$

Indeed the right side is $4 - 4b_1 + 2(b_+ + b_-) + 2(b_+ - b_-) = 4(1 - b_1 + b_+)$. Substitution gives $d = (c^2 - 2\chi - 3\sigma)/4$. In the simply connected case $b_1 = 0$, the real contribution from the gauge and self-dual form complex is $-1 - b_+$; in the based quotient it is $-b_+$. This identifies the two dimension calculations in the original transversality discussion and makes clear why the based dimension exceeds the ordinary one by one.

Theorem 3.6 (Generic smoothness [12, 17]). *For a fixed closed oriented Spin^c four-manifold, a residual set of smooth self-dual perturbations makes the irreducible moduli space $\mathcal{M}_{\mathfrak{s}, \eta}^*$ a smooth manifold of dimension $d(\mathfrak{s})$ when it is nonempty. A homology orientation, namely an orientation of $H^1(M; \mathbb{R}) \oplus H_+^2(M; \mathbb{R})$, orients it. If $b_+ > 0$, a generic perturbation also excludes reducibles, so the full moduli space is a compact oriented smooth manifold of that dimension.*

Sketch of proof. Apply the Sard–Smale theorem to the universal equations with η varying. At an irreducible pair, unique continuation for the Dirac equation and variation of η give the required surjectivity of the universal linearization. The index calculation is (1); the complex orientation of D_A^+ together with the chosen homology orientation gives the orientation. A reducible satisfies $F_A^+ = \eta$. By the Hodge calculation of Section 1.3, the set of η for which that equation has a solution has codimension b_+ , so it can be avoided when $b_+ > 0$. Sard–Smale, the slice theorem, and unique continuation are used here as external results, not proved in this report.

The variation of the equations at (A, ψ) in a connection direction a and spinor direction ϕ has the form

$$(a, \phi) \mapsto \left(D_A^+ \phi + \frac{i}{2} \rho(a) \psi, i d^+ a - dq_\psi(\phi) \right),$$

where dq_ψ is the real differential of the quadratic map. Infinitesimal gauge directions lie in its kernel and are removed by a slice condition. The derivative in the perturbation direction adds $-\dot{\eta}$ to the second component. Surjectivity of the universal operator at irreducibles, followed by Sard–Smale projection to the space of perturbations, is the precise mechanism behind the word “generic.” A zero expected dimension then yields a finite signed set by compactness; negative expected dimension gives an empty irreducible moduli space. The theorem does not assert regularity at the reducible when $b_+ = 0$. $\square$

3.3 Proof of Donaldson’s Theorem

The intersection form of a closed oriented four-manifold is the cup-product pairing on the free part of $H^2(M; \mathbb{Z})$, evaluated on $[M]$. It is unimodular by Poincaré duality. Donaldson originally obtained the following restriction using nonabelian gauge theory [2]; a Seiberg–Witten argument for it is given in [7].

Theorem 3.7 (Donaldson diagonalization theorem). *If M is a closed, connected, oriented, smooth, simply connected four-manifold with negative definite intersection form of rank n , then its form is isomorphic over $\mathbb{Z}$ to $-I_n$. By reversing orientation, a positive definite form is I_n .*

Sketch of proof following [7]. Suppose the negative definite unimodular lattice Q_M is not diagonal. Elkies’s lattice theorem [3] supplies a characteristic class $c \in H^2(M; \mathbb{Z})$ with

$$c^2 > -n, \quad c^2 \equiv -n \pmod{8}.$$

A characteristic class reduces to $w_2(TM)$, so choose a Spin^c structure with determinant class c . Since $b_1 = b_+ = 0$ and $\sigma = -n$, its Dirac index and expected dimension are

$$m = \text{ind}_{\mathbb{C}} D_A^+ = \frac{c^2 + n}{8} > 0, \quad d(\mathfrak{s}) = 2m - 1.$$

Thus the based moduli space has expected dimension $2m$. For any self-dual perturbation there is exactly one reducible orbit: the equation $F_A^+ = \eta$ has a solution because $b_+ = 0$, and two such connections differ by a gauge because $H^1(M; \mathbb{R}) = 0$.

The congruence $c^2 \equiv -n \pmod{8}$ makes m an integer; the strict inequality makes it positive. This choice of a characteristic vector is the extra lattice-theoretic ingredient needed for an *arbitrary* definite form. If the form is even, $c = 0$ already works and one can see the original calculation $m = n/8$. The reducible is unique in the unbased quotient, but it is a fixed point of the residual circle acting on the based quotient. It is precisely here that a naive application of the ordinary transversality theorem would fail.

The generic-smoothness theorem above applies away from the reducible. It does *not* assert smoothness at that point. The additional equivariant Dirac perturbation and regularity argument of [7, Secs. 4–5] arrange a smooth based moduli space near the reducible while retaining compactness. At that point the tangent space is $\ker D_A^+$, now a complex vector space of dimension m because the perturbed Dirac operator is surjective. The residual S^1 acts by scalar multiplication. Consequently the unbased moduli space locally is $\mathbb{C}^m/S^1$, a cone with link $\mathbb{CP}^{m-1}$. Remove a small cone neighborhood. Compactness leaves a $(2m - 1)$ -manifold V whose only boundary is this projective space.

Indeed a nonzero vector in $\mathbb{C}^m$ has a positive radius and a direction on S^{2m-1} . The circle quotient removes its phase, leaving $(0, \varepsilon) \times \mathbb{CP}^{m-1}$; adding the zero vector collapses the projective space at radius zero. This is the cone description. Its dimension is $1 + 2(m - 1) = 2m - 1$, as expected from the index. The boundary of the truncated cone is one copy of $\mathbb{CP}^{m-1}$, while compactness rules out additional boundary from escaping sequences of solutions. The local regularity and global compactness here are theorem-level inputs, not consequences of this elementary quotient calculation alone.

Over the irreducible part the based quotient is a principal S^1 -bundle. Its restriction to $\partial V = \mathbb{CP}^{m-1}$ is the Hopf fibration $S^{2m-1} \rightarrow \mathbb{CP}^{m-1}$, up to reversing the circle action. If u is the first Chern class of its associated line bundle, then

The bundle from the original argument can be displayed explicitly. On the irreducible configuration space put $\mathbb{E} = \tilde{\mathcal{B}}^* \times_{S^1} \mathbb{C}$; its unit sphere bundle is $S(\mathbb{E}) \cong \tilde{\mathcal{B}}^*$. The inclusions of the moduli spaces respect the residual circle action:

$$\begin{array}{ccc} \widetilde{\mathcal{M}}_{s,\eta}^* & \longrightarrow & \widetilde{\mathcal{B}}^* \\ \downarrow & & \downarrow \\ \mathcal{M}_{s,\eta}^* & \longrightarrow & \mathcal{B}^*. \end{array}$$

Restricting $S(\mathbb{E})$ over the projective link gives $S^{2m-1} \rightarrow \mathbb{CP}^{m-1}$. Write $u = c_1(\mathbb{E}|_V)$. Up to sign its restriction to the link generates $H^2(\mathbb{CP}^{m-1}; \mathbb{Z})$, whose ring is $\mathbb{Z}[u]/(u^m)$. Consequently

$$|\langle u^{m-1}, [\mathbb{CP}^{m-1}] \rangle| = 1.$$

But u extends over V , so its evaluation on the oriented boundary vanishes. In homological terms, $[\partial V]$ maps to zero in $H_{2m-2}(V; \mathbb{Z})$ because it is the boundary of its relative fundamental class; pairing with u^{m-1} would give zero. For $m = 1$ this says that a compact oriented one-manifold cannot have a single boundary point. We have a contradiction. The cited argument supplies the equivariant perturbation and local smooth model essential to the sketch. $\square$

The case of an even negative definite form is especially transparent: $c = 0$ is characteristic and $m = n/8$. The calculation explains the original cone picture but does not assume that the trivial determinant connection has an automatically surjective Dirac operator.

4 Some Applications

4.1 Further Results and Seiberg–Witten Invariants

Assume $b_+(M) > 1$ and fix a homology orientation. Generic perturbations then give compact smooth moduli spaces with no reducibles, and one-parameter families of perturbations provide the cobordisms needed for invariance [17, 12]. In expected dimension zero, the invariant is a signed count. Higher even dimensions require a cohomology class on the configuration quotient.

To obtain an integer, the zero-dimensional moduli space must first be oriented. The orientation of $H^1(M; \mathbb{R}) \oplus H_+^2(M; \mathbb{R})$ fixes the real part of the determinant line of the deformation complex, while the Dirac part has its complex orientation. A generic path between two metrics and perturbations produces a one-dimensional parameterized moduli space when the expected dimension is zero. For $b_+ > 1$, its reducible locus can be avoided along a generic one-parameter path; compactness makes it an oriented cobordism between the two zero-dimensional spaces. The signed boundary count is zero, explaining metric and perturbation independence. If $b_+ = 1$, the reducible condition has codimension one, so a generic path may cross it and a wall-crossing phenomenon appears.

The principal S^1 -bundle $\tilde{\mathcal{B}}^* \rightarrow \mathcal{B}^*$ from Section 3.1 determines an associated complex line bundle $\mathbb{E} \rightarrow \mathcal{B}^*$. Fix the sign convention $\mu(\text{pt}) = c_1(\mathbb{E})$ (reversing the circle-action convention changes this sign). The same class restricts to every irreducible moduli space.

The class $\mu(\text{pt})$ has cohomological degree two. Thus if the moduli space has real dimension $d = 2r$, its r th power has top degree and can be paired with the fundamental class. The principal bundle here is the same based-to-unbased circle bundle whose restriction over the link of a reducible was the Hopf bundle in the preceding section. In positive dimension a “count of solutions” without such a degree-matching insertion has no numerical meaning.

Definition 4.1. For $d = d(\mathfrak{s}) \geq 0$ even and a generic perturbation, define the numerical Seiberg–Witten invariant used here by

$$\text{SW}_M(\mathfrak{s}) = \langle \mu(\text{pt})^{d/2}, [\mathcal{M}_{\mathfrak{s}, \eta}] \rangle.$$

For $d < 0$ or odd d , set this numerical invariant to zero. When $d = 0$, the formula is the signed count $\sum_{[A, \psi]} \text{sign}[A, \psi]$ of the finitely many regular solutions. More general Seiberg–Witten invariants allow insertions from $H_1(M)$; we use only the numerical version above.

The value is independent of the generic metric and perturbation when $b_+ > 1$, up to the stipulated homology orientation. With $b_+ = 1$ a chamber choice is needed, and the same metric-independent statement is false. The point insertion is why a positive-dimensional moduli space is not counted simply by its number of points.

For $d = 0$, $\mu(\text{pt})^0 = 1$, so pairing with the oriented fundamental class is exactly the sum of local orientation signs. For $d = 2$, the same definition integrates $c_1(\mathbb{E})$ over the moduli surface. The full Seiberg–Witten polynomial can also use degree-one classes associated to loops in M ; the numerical invariant chosen in this report is sufficient for the applications below. A moduli space can be nonempty while this pairing vanishes, so being basic is stronger than mere existence of solutions.

There are several descriptions of its domain. First, SW_M is a function on isomorphism classes of Spin^c structures. Second, $c_1(\mathcal{L})$ is characteristic, that is, its reduction modulo two equals $w_2(TM)$. If $H^2(M; \mathbb{Z})$ has no two-torsion, different Spin^c structures have different determinant classes, so one may write $\text{SW}_M(c)$ for characteristic c . If two-torsion is present, the same c can correspond to different structures, and that notation loses information. Finally, after choosing a reference $\mathfrak{s}_0$, twisting by lines identifies the set of Spin^c structures as an affine space over $H^2(M; \mathbb{Z})$:

$$\mathfrak{s} = \mathfrak{s}_0 \otimes E, \quad c_1(\det \mathfrak{s}) = c_1(\det \mathfrak{s}_0) + 2c_1(E).$$

Definition 4.2. A Spin^c structure $\mathfrak{s}$ is *basic* if $\text{SW}_M(\mathfrak{s}) \neq 0$. When the determinant class determines the structure, $c_1(\det \mathfrak{s})$ is also called a basic class. The manifold has *Seiberg–Witten simple type* if $\text{SW}_M(\mathfrak{s}) \neq 0$ implies $d(\mathfrak{s}) = 0$.

For a fixed metric and perturbation the spinor estimate bounds the self-dual curvature. The condition $d(\mathfrak{s}) \geq 0$ gives a uniform lower bound on $c_1(\det \mathfrak{s})^2$. Together with the curvature identity used in the compactness proof, this bounds the full curvature in L^2 . The discrete integral lattice then gives only finitely many possible determinant classes; there are also only finitely many lifts with a given determinant on a compact manifold [17, 12]. Simple type is a condition on *nonzero invariants*, not a claim that all moduli spaces have dimension zero.

The finiteness argument should be read with the order of its bounds in mind. The maximum principle gives a uniform estimate for F_A^+ across all structures with a solution, because the bound depends on the fixed metric and perturbation. If $d(\mathfrak{s}) \geq 0$, the dimension formula says $c_1(\det \mathfrak{s})^2 \geq 2\chi + 3\sigma$. In the curvature identity, this bounds the anti-self-dual part as well. Hodge theory turns the resulting bound on the harmonic representative of $c_1(\det \mathfrak{s})$ into a bound in the finite-dimensional real vector space $H^2(M; \mathbb{R})$. A bounded region contains only finitely many points of its integral lattice modulo torsion, and the torsion subgroup of $H^2(M; \mathbb{Z})$ is finite. This is why the generating series in the knot-surgery section is a finite sum.

Theorem 4.3 (Conjugation symmetry [17]). *For $b_+ > 1$, conjugation of Spin^c structures satisfies*

$$\text{SW}_M(\bar{\mathfrak{s}}) = \pm \text{SW}_M(\mathfrak{s}), \quad c_1(\det \bar{\mathfrak{s}}) = -c_1(\det \mathfrak{s}).$$

The sign depends on orientation conventions. Thus basic determinant classes occur in opposite pairs whenever they distinguish the structures.

Conjugation changes the determinant line to its complex conjugate and sends the first Chern class to its negative. It also conjugates the spinor representation and pairs corresponding moduli spaces. The induced orientation can change by a sign depending on the chosen homology orientation and index parity, so the unqualified equality of the two integer values would require an additional convention. The invariant’s *nonvanishing*, which is what matters in the adjunction and symplectic applications, is symmetric under this operation.

Theorem 4.4 (Positive scalar curvature vanishing [17]). *If a closed oriented smooth four-manifold has $b_+ > 1$ and admits a metric of positive scalar curvature, all its numerical Seiberg–Witten invariants vanish.*

Sketch of proof. The unperturbed spinor bound forces every solution for such a metric to be reducible. Positive scalar curvature has a positive minimum, so the same bound gives $\psi = 0$ for sufficiently small perturbations. Choose one that excludes reducibles, possible since $b_+ > 1$. Its moduli space is empty, and invariance then gives the assertion. Compactness and generic-perturbation invariance are the cited analytic results.

Concretely, if $s \geq s_0 > 0$ and $2\|\rho(\eta)\|_{L^\infty} < s_0$, the bound in the lemma would force $|\psi|^2 < 0$ at a positive maximum; hence $\psi = 0$. Reducible solutions would have $F_A^+ = \eta$. Their possible perturbations form the affine subspace of codimension b_+ described in Hodge theory. Choose a sufficiently small η outside it. Then no solution remains, so the invariant computed with this perturbation is zero. The cobordism argument above transports that value to all generic choices when $b_+ > 1$. $\square$

4.2 The Minimal Genus of Embedded Surfaces

Given $x \in H_2(M; \mathbb{Z})$, one may ask for the smallest genus of a connected smoothly embedded oriented surface representing x . The self-intersection of a surface Σ is $[\Sigma]^2 = \langle \text{PD}[\Sigma]^2, [M] \rangle$, also the Euler number of its oriented normal bundle.

A classical obstruction concerns spheres representing characteristic classes, before the stronger gauge-theoretic genus bounds.

Theorem 4.5 (Kervaire–Milnor [8]). *Let M be closed, oriented, and smooth, and let $\Sigma \subset M$ be a smoothly embedded two-sphere whose integral homology class is characteristic for the intersection pairing. Then*

$$\sigma(M) - [\Sigma]^2 \equiv 0 \pmod{16}.$$

Here $[\Sigma]$ is characteristic when $[\Sigma] \cdot x \equiv x^2 \pmod{2}$ for all $x \in H_2(M; \mathbb{Z})/\text{torsion}$. This congruence constrains the sphere's possible normal Euler number. Seiberg–Witten theory also gives inequalities involving arbitrary basic classes.

The hypothesis is about the homology class, not just the topology of the embedded sphere. For example, a sphere whose class is not characteristic does not fall under this congruence. If $[\Sigma]^2$ is the Euler number of its normal bundle, the displayed relation compares that local embedding datum with the signature of the whole manifold. A genus inequality instead bounds how complicated *any* surface representing a fixed nontrivial class must be, once a Seiberg–Witten basic class is known.

Theorem 4.6 (Adjunction inequality [9, 12]). *Let M be closed, connected, oriented, and smooth with $b_+(M) > 1$, and let $\Sigma \subset M$ be a connected, smoothly embedded, oriented surface of genus $g \geq 1$ with $[\Sigma] \neq 0$ and $[\Sigma]^2 \geq 0$. For every basic determinant class κ ,*

$$2g - 2 \geq [\Sigma]^2 + |\langle \kappa, [\Sigma] \rangle|, \quad \text{equivalently} \quad \chi(\Sigma) + [\Sigma]^2 \leq -|\langle \kappa, [\Sigma] \rangle|.$$

Sketch of proof. For $[\Sigma]^2 = 0$, the oriented normal bundle is trivial. A tubular neighborhood is $\Sigma \times D^2$, and its boundary is $\Sigma \times S^1$; the long neck used in the argument is $\Sigma \times S^1 \times [0, T]$, not the tubular neighborhood itself. The Seiberg–Witten estimates on a stretched neck, together with Gauss–Bonnet $\int_{\Sigma} K dA = 2\pi\chi(\Sigma)$ and the nonvanishing of the invariant for κ , imply $|\kappa \cdot \Sigma| \leq 2g - 2$. Establishing the required limiting solutions and estimates is the substantial gluing and analysis in the cited adjunction results. There is no general pointwise inequality $|s| \geq 2\sqrt{2}|F_A^+|$ for each solution; the original attempted deduction from that inequality is therefore invalid.

The square-zero case is the analytic core. Because the normal bundle is trivial, one may insert a cylinder with cross-section $\Sigma \times S^1$ and let its length T grow. The Seiberg–Witten compactness and gluing analysis relates a nonzero invariant on M to solutions on the stretched pieces and to the induced three-dimensional equations on the cross-section. The latter impose a degree bound on the line data over Σ . Gauss–Bonnet converts its curvature contribution to $\chi(\Sigma) = 2 - 2g$, giving $|\kappa \cdot \Sigma| \leq -\chi(\Sigma)$. Both signs occur because conjugation replaces κ by $-\kappa$. The original proposed formulas with unspecified ρ and $O(1)$ terms do not determine a valid limit or even the sign of $\kappa \cdot \Sigma$; the cited neck-stretching theorem supplies the missing estimates and compactness rather than those formulas being taken as a proof.

If $n = [\Sigma]^2 > 0$, blow up n distinct points of Σ to obtain its proper transform $\tilde{\Sigma} \subset M \# n \overline{\mathbb{CP}}^2$ with $[\tilde{\Sigma}]^2 = 0$ and the same genus. Its homology class is $[\tilde{\Sigma}] = [\Sigma] - \sum_j e_j$, where each exceptional sphere e_j has square -1 . Consequently $[\tilde{\Sigma}]^2 = n - n = 0$. The Seiberg–Witten blow-up formula gives basic classes $\kappa + \sum_j \epsilon_j E_j$, where $E_j = \text{PD}(e_j)$ and $\epsilon_j = \pm 1$. Their pairing with $[\tilde{\Sigma}]$ is $\kappa \cdot \Sigma + \sum_j \epsilon_j$, up to the chosen sign for E_j . Choosing signs so its absolute value is $|\kappa \cdot \Sigma| + n$ and applying the square-zero inequality gives the stated bound. The blow-up formula is an external theorem [12]. $\square$

For surfaces of negative square, variants require additional hypotheses and specialized adjunction results. The theorem above is the precise nonnegative-square claim used here.

Corollary 4.7 (Nonnegative spheres [9]). *If $b_+(M) > 1$ and M contains a homologically nontrivial embedded sphere S with $S^2 \geq 0$, then $\text{SW}_M \equiv 0$.*

This is the genus-zero vanishing result from the same gauge-theoretic theory, rather than a literal substitution of $g = 0$ into Theorem 4.6, which assumed $g \geq 1$.

The corollary has a clear geometric meaning: on a manifold with a nonzero Seiberg–Witten invariant and $b_+ > 1$, an essential sphere must have negative self-intersection. It is stronger in this setting than a restriction only on characteristic spheres, because it applies to any nontrivial sphere of nonnegative square. We use the separately cited genus-zero result rather than extending the hypotheses of the positive-genus inequality without proof.

4.3 Fintushel–Stern Knot Surgery

Knot surgery changes a smooth four-manifold in a torus neighborhood while retaining much of its topology. A useful geometric hypothesis on the torus ensures that the Seiberg–Witten calculation has a particularly simple form [4].

Definition 4.8. Let $T \subset M$ be an oriented, homologically essential, smoothly embedded torus with $T^2 = 0$. We call T *cuspidally embedded* (the near-cusp hypothesis in the original discussion) when it is a regular torus fiber in an embedded cusp neighborhood. Such a neighborhood has two vanishing cycles on the fiber, represented by curves meeting once; their thimbles have relative framing -1 . This additional neighborhood condition is stronger than $T^2 = 0$ alone.

If $b_+ > 1$ and M has a basic class κ , Theorem 4.6 applied to T gives $\langle \kappa, [T] \rangle = 0$. Thus all basic classes are orthogonal to a homologically essential square-zero torus, whether or not a cusp neighborhood is present.

Here the genus of T is one and its self-intersection is zero, so the two sides of the adjunction inequality are $2g(T) - 2 = 0$ and $|\kappa \cdot T|$. Nonnegativity forces the pairing to vanish. This is why powers of $e^{\text{PD}[T]}$ can shift basic classes in a knot-surgery formula without changing their pairing with the torus itself. The near-cusp or cusp condition plays a different role: it supports the gluing calculation and, under the stated complement conditions, the topology of the fiber sum.

To encode the invariants, assume for the moment that the basic determinant classes can be used without a two-torsion ambiguity. For $\alpha \in H^2(M; \mathbb{Z})/\text{torsion}$ introduce formal symbols e^α with $e^\alpha e^\beta = e^{\alpha+\beta}$.

Definition 4.9 (Seiberg–Witten series). The associated finite element of the integral group ring is

$$\mathcal{SW}_M = \sum_{\mathfrak{s} \text{ basic}} \text{SW}_M(\mathfrak{s}) e^{c_1(\det \mathfrak{s})}.$$

When distinct basic structures have the same determinant modulo torsion, their coefficients are combined in this series.

An orientation of M and a homology orientation fix the overall signs. If $[T]$ is a homology class, write $t = e^{\text{PD}[T]}$. This distinguishes the torus from its Poincaré dual in formulas.

Suppose M and M' contain square-zero tori T, T' . Their oriented tubular neighborhoods are $T^2 \times D^2$ and $T'^2 \times D^2$. Removing their interiors leaves boundary three-tori. An orientation-reversing

boundary diffeomorphism identifying the circle meridians produces the fiber sum $M \#_{T=T'} M'$. The boundary identification, and in general possible rim-torus classes, are part of the construction.

The zero self-intersection trivializes the oriented normal disk bundle, which is why its boundary is $T^2 \times S^1 = T^3$. Gluing the complements along an orientation-reversing map gives a closed oriented four-manifold. The map must specify what happens to a normal circle as well as to the two cycles in T ; a different choice can alter fundamental group and homology. In the simple hypotheses of the theorem below, the cohomology classes from the two torus fibers are identified and the group-ring variable can be denoted by the same t on both sides.

Theorem 4.10 (Torus fiber-sum formula [4]). *Suppose M, M' are closed simply connected smooth four-manifolds with $b_+ > 1$, containing primitive cusp-embedded square-zero tori T, T' with simply connected complements. For the standard fiber-sum gluing and the induced identifications of the relevant cohomology classes, the Seiberg–Witten series satisfies*

$$\mathcal{SW}_{M \#_{T=T'} M'} = \mathcal{SW}_M \mathcal{SW}_{M'} (t - t^{-1})^2, \quad t = e^{\text{PD}[T]} = e^{\text{PD}[T']}.$$

Signs are fixed consistently by the homology orientations. The displayed formula is the usual form when no extra rim-torus variables occur.

The two factors $(t - t^{-1})$ in the original formula refer to the two tori, which are identified in the sum. Without the gluing and complement conditions a formula in this one-variable form need not hold.

The formula is a product in a group ring, after passing the basic classes from the two summands to the fiber sum. For example, if terms ae^α and be^β occur in the two original series, they contribute

$$ab e^{\alpha+\beta} (t^2 - 2 + t^{-2})$$

to the resulting formal expression before like terms are combined. This calculation shows why the two factors in the original display cannot be interpreted as independent free variables after the tori have been identified. The theorem is a gluing result for monopole invariants; the algebraic multiplication is its consequence, not a replacement for the analytic gluing theorem.

For a knot $K \subset S^3$, let νK be an open tubular neighborhood. Its exterior $S^3 \setminus \nu K$ has torus boundary; $(S^3 \setminus \nu K) \times S^1$ therefore has three-torus boundary and the homology of $T^2 \times D^2$. Remove $\nu T \cong T^2 \times D^2$ and identify the boundaries so that the meridian of T maps to the preferred longitude of K (with the other circles matched in the standard way). Define

$$M_K = (M \setminus \nu T) \cup_\varphi ((S^3 \setminus \nu K) \times S^1).$$

The specified gluing is needed: matching arbitrary boundary circles can change the result. Mayer–Vietoris shows that this operation preserves integral homology and the intersection form under the standard hypotheses. If M and $M \setminus \nu T$ are simply connected, van Kampen gives $\pi_1(M_K) = 1$ for the standard gluing. Freedman’s topological classification then identifies M_K and M up to homeomorphism (with the corresponding Kirby–Siebenmann data) [5, 4].

The preferred longitude of a knot has zero class in $H_1(S^3 \setminus \nu K; \mathbb{Z})$, whereas its meridian generates that homology. Matching the torus meridian to this longitude makes the replacement homologically behave like the removed $T^2 \times D^2$. The Mayer–Vietoris sequence then compares the pieces and their common T^3 boundary to yield the homology claim. Simple connectivity of the complement is needed to kill the remaining loops in the van Kampen calculation. Thus the homeomorphism assertion has topological hypotheses beyond the bare statement that the two pieces have the same homology.

Theorem 4.11 (Fintushel–Stern knot-surgery formula [4]). *Let M be closed, simply connected, and smooth with $b_+(M) > 1$, and let $T \subset M$ be a homologically essential cusp-embedded torus of square zero with simply connected complement. For the standard knot surgery above,*

$$\mathcal{SW}_{M_K} = \mathcal{SW}_M \Delta_K(t^2), \quad t = e^{\text{PD}[T]},$$

where $\Delta_K(z) = \Delta_K(z^{-1})$ and $\Delta_K(1) = 1$ normalize the symmetrized Alexander polynomial.

Thus knots with distinct normalized Alexander polynomials can produce different Seiberg–Witten series and hence nondiffeomorphic smooth structures, provided the product series distinguish them. Knots with the same Alexander polynomial need not be distinguished by this invariant. Under the complement hypothesis the resulting manifolds are nevertheless homeomorphic to M .

The argument for a smooth-structure distinction uses diffeomorphism invariance of the series, with the induced action on cohomology taken into account. If the starting series is nonzero and the resulting series cannot be identified under such an action, no diffeomorphism exists. The formula does not claim that every pair of different knots is detected, nor that the Alexander polynomial uniquely determines the knot. It supplies a calculable source of examples while the preceding topological argument explains how their homeomorphism type can stay fixed.

4.4 Relation to Symplectic Topology

Seiberg–Witten theory is closely related to symplectic four-manifolds. Taubes’s results connect monopole invariants to counts of pseudo-holomorphic subvarieties [13, 16].

Definition 4.12. A symplectic form on a smooth four-manifold is a closed, nondegenerate real two-form ω . We orient M by requiring $\omega \wedge \omega > 0$. In this orientation, for closed M ,

$$\langle [\omega]^2, [M] \rangle = \int_M \omega \wedge \omega > 0,$$

so $b_+(M) \geq 1$.

Nondegeneracy means the linear map $v \mapsto \iota_v \omega$ from $T_p M$ to $T_p^* M$ has zero kernel at every point. In dimension four it is equivalent to $\omega \wedge \omega$ being a nowhere-zero four-form; choosing the symplectic orientation makes the latter positive. Closedness makes $[\omega]$ a cohomology class, and the integral shows it has positive square for the intersection form. Thus symplectic geometry supplies precisely the kind of positive subspace that already appeared in Hodge theory. The conclusion $b_+ \geq 1$ alone does not ensure $b_+ > 1$, so the chamber-free Seiberg–Witten statements below retain the stronger hypothesis.

Definition 4.13. An almost complex structure is a smooth bundle map $J : TM \rightarrow TM$ satisfying $J^2 = -\text{id}$. It is compatible with ω if $g(v, w) = \omega(v, Jw)$ is a Riemannian metric; equivalently $\omega(Jv, Jw) = \omega(v, w)$ and $\omega(v, Jv) > 0$ for $v \neq 0$. The triple (M, ω, J) is then almost Kähler.

The first condition says that J turns each real tangent space into a complex vector space; compatibility makes ω positive in the J directions. One need not assume that J is integrable. A J -holomorphic curve is an immersed or suitably singular surface whose tangent planes are J -invariant, with a parametrization u satisfying $du \circ j = J \circ du$ where it is smooth. Such curves are the objects counted, with appropriate weights, on the Gromov side of Taubes’s theorem. Compatible J exist and form a contractible space. It determines a canonical Spin^c structure $\mathfrak{s}_J$ with

$$W_J^+ \cong \Lambda^{0,0} T^* M \oplus \Lambda^{0,2} T^* M, \quad W_J^- \cong \Lambda^{0,1} T^* M, \quad \det W_J^+ \cong K_M^{-1},$$

where $K_M = \Lambda^{2,0}T^*M$ is the canonical line bundle. Put $K = c_1(K_M) = -c_1(TM, J)$. This convention distinguishes the canonical class K from its negative $-K = c_1(K_M^{-1})$. Every Spin^c structure is $\mathfrak{s}_J \otimes E$ for a line bundle E , with $e = c_1(E)$, and

$$c_1(\det(\mathfrak{s}_J \otimes E)) = -K + 2e.$$

In this notation the original report's K_M^* is the cohomology class $-K$ if the star denotes the anti-canonical line, rather than a new independent canonical class. The determinant of $W_J^+ = \Lambda^{0,0} \oplus \Lambda^{0,2}$ is $\Lambda^{0,2}$, which as a complex line is isomorphic to K_M^{-1} using the compatible metric. Twisting both spinor summands by E multiplies the determinant by E^2 , explaining the coefficient $2e$. These identifications connect the symplectic construction directly to the general twisting rule from the spin geometry section. For an almost complex four-manifold $K^2 = 2\chi(M) + 3\sigma(M)$. The dimension formula (1) becomes

$$d(\mathfrak{s}_J \otimes E) = \frac{(-K + 2e)^2 - K^2}{4} = e^2 - K \cdot e.$$

This is the original dimension calculation with the sign of the canonical class made explicit.

For clarity, expand the numerator:

$$(-K + 2e)^2 - K^2 = K^2 - 4K \cdot e + 4e^2 - K^2 = 4(e^2 - K \cdot e).$$

In particular the canonical structure has $e = 0$ and expected dimension zero, so its nonzero Taubes invariant is indeed an ordinary signed count. The formula also shows how the geometry of a potential curve class $\text{PD}(e)$ affects the expected dimension of the corresponding monopole moduli space.

Theorem 4.14 (Taubes [13, 16]). *If (M, ω) is a closed symplectic four-manifold with $b_+(M) > 1$, then it has Seiberg–Witten simple type. With the symplectic homology orientation, $\text{SW}_M(\mathfrak{s}_J) = \pm 1$, so $-K$ is a basic determinant class; conjugation also makes K basic. Moreover, if $\text{SW}_M(\mathfrak{s}_J \otimes E) \neq 0$, then $\text{PD}(e)$ is represented by a (possibly disconnected and multiply covered) J -holomorphic subvariety for a suitable generic compatible J .*

The refined $\text{SW} = \text{Gr}$ statement identifies $\text{SW}_M(\mathfrak{s}_J \otimes E)$, up to fixed orientation conventions, with Taubes's Gromov invariant $\text{Gr}(\text{PD}(e))$ [14, 16, 15]. The latter is a weighted signed count with provisions for multiply covered tori and disconnected curves; it is not in general the cardinality of a finite set of individual J -holomorphic curves. This precision matters even when the Seiberg–Witten moduli space is zero-dimensional. It also explains why the canonical class, genus bounds, and smooth structures discussed above are connected to symplectic geometry.

The existence part and the equality part have different logical strengths. Nonvanishing of the invariant for $\mathfrak{s}_J \otimes E$ forces a J -holomorphic representative of $\text{PD}(e)$ in the Taubes correspondence; it does not say that there is a unique smooth embedded curve. The equality $\text{SW} = \text{Gr}$ compares two signed invariants defined with compatible orientation and multiplicity conventions. A union of components, including multiple covers of square-zero tori, contributes according to the Gromov invariant's rules rather than one per visible curve. This is why the original formula using a bare cardinality could not be retained. For the canonical structure $e = 0$, the nonzero invariant gives the basic class $-K$; conjugation supplies K . Those classes can then be inserted in the adjunction inequality to constrain genera of surfaces in symplectic four-manifolds.

References

- [1] M. F. Atiyah and I. M. Singer, The index of elliptic operators: III, *Ann. of Math. (2)* **87** (1968), 546–604.
- [2] S. K. Donaldson, An application of gauge theory to four-dimensional topology, *J. Differential Geom.* **18** (1983), 279–315.
- [3] N. D. Elkies, A characterization of the $\mathbb{Z}^n$ lattice, *Math. Res. Lett.* **2** (1995), 321–326.
- [4] R. Fintushel and R. J. Stern, Knots, links, and 4-manifolds, *Invent. Math.* **134** (1998), 363–400.
- [5] M. H. Freedman, The topology of four-dimensional manifolds, *J. Differential Geom.* **17** (1982), 357–453.
- [6] J. Jost, *Riemannian Geometry and Geometric Analysis*, 7th ed., Springer, Cham, 2017.
- [7] M. G. Katz, A proof via the Seiberg–Witten moduli space of Donaldson’s theorem on smooth 4-manifolds with definite intersection forms, *R. C. P. 25, Prépubl. IRMA Strasbourg* **47** (1995), 269–274.
- [8] M. A. Kervaire and J. W. Milnor, On 2-spheres in 4-manifolds, *Proc. Natl. Acad. Sci. USA* **47** (1961), 1651–1657.
- [9] P. B. Kronheimer and T. S. Mrowka, The genus of embedded surfaces in the projective plane, *Math. Res. Lett.* **1** (1994), 797–808.
- [10] H. B. Lawson, Jr. and M.-L. Michelsohn, *Spin Geometry*, Princeton University Press, Princeton, NJ, 1989.
- [11] J. Milnor and J. Stasheff, *Characteristic Classes*, Princeton University Press, Princeton, NJ, 1974.
- [12] J. W. Morgan, *The Seiberg–Witten Equations and Applications to the Topology of Smooth Four-Manifolds*, Princeton University Press, Princeton, NJ, 1996.
- [13] C. H. Taubes, The Seiberg–Witten invariants and symplectic forms, *Math. Res. Lett.* **1** (1994), 809–822.
- [14] C. H. Taubes, Counting pseudo-holomorphic submanifolds in dimension 4, *J. Differential Geom.* **44** (1996), 818–893.
- [15] C. H. Taubes, $\text{Gr} \Rightarrow \text{SW}$: From pseudo-holomorphic curves to Seiberg–Witten solutions, *J. Differential Geom.* **51** (1999), 203–334.
- [16] C. H. Taubes, $\text{SW} \Rightarrow \text{Gr}$: From the Seiberg–Witten equations to pseudo-holomorphic curves, *J. Amer. Math. Soc.* **9** (1996), 845–918.
- [17] E. Witten, Monopoles and four-manifolds, *Math. Res. Lett.* **1** (1994), 769–796.